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Multipurpose electric field shield for wireless power transfer systems

Abstract

A wireless power transfer system is disclosed. The system includes a wireless power transmitter including a transmitter coil and a transmitter electric field shield disposed over the transmitter coil. The transmitter electric field shield includes a first printed circuit board (PCB) and a second PCB. Each of the first and second PCBs includes a number of apertures. The apertures of the first PCB do not overlap with the apertures of the second PCB. The system further includes a wireless power receiver including a receiver coil and a receiver electric field shield disposed over or underneath the receiver coil. The receiver electric field shield includes a PCB having a first layer and a second layer connected to one another. The first layer of the PCB mates with the receiver coil and the second layer of the PCB mates with an interface surface of the wireless power receiver.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/446,457 filed on Feb. 17, 2023, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure generally relates to wireless power transfer systems, and more specifically, to electric field shielding for wireless power transfer systems.

BACKGROUND

(2) Wireless devices may be configured to utilize various wireless charging components to recharge batteries and other power storage devices. Accordingly, such wireless devices may have associated wireless power transfer systems (e.g., charging stations), and such devices and systems may have transmitters and receivers including, among other things, inductive elements configured for charging operations. Moreover, wireless devices and their associated wireless power transfer systems may be capable of multiple different charging modes.

(3) A high power wireless power transfer system, which includes a transmitter and a receiver, generally has significant power loss contributed by transmitter and receiver coils. Thus, in a typical high power wireless power transfer system, a cooling fan is required for cooling the interface surface (i.e., mating surface of transmitter and receiver coils). For automotive applications, where the transmitter coil is enclosed, the cooling of the interface surface is imperative. Further, the high power wireless system also requires an electric field shield for the transmitter and receiver coils. The purpose of the shield is to shield the electric field and let pass of the magnetic field. The electric field shielding reduces the electromagnetic interference (EMI). The EMI requirements are

more stringent for automotive applications and using a shield to meet the EMI requirements is unavoidable.

(4) While conventional wireless power transfer systems address EMI issues by using shielding to reduce EMI, such shielding degrades thermal performance of the systems as shield covers the interface surface, thereby blocking air flow to transmitter and receiver coils.

(5) In a typical wireless system, the transmitter generally includes a cooling fan to cool the interface surface and the receiver generally includes functions to allow the fan air to cool the receiver. Further, reliable foreign object detection (FOD) is generally a challenge particularly when foreign object (FO) resistance is placed in a corner of the interface surface so that it can escape the Q factor and power loss detection method but is large enough to cause heat.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an example of a system for wireless charging in accordance with various embodiments.

(2) FIG. 2 is a block diagram illustrating another example of a system for wireless charging in accordance with various embodiments.

(3) FIGS. 3A-3B are diagrams illustrating an example of a transmitter electric field shield in accordance with various embodiments.

(4) FIG. 4 is a diagram illustrating an example of a receiver electric field shield in accordance with various embodiments.

(5) FIG. 5 is a diagram illustrating a sectional view of an electric field shielding system for interface surface cooling and shielding in accordance with various embodiments.

(6) FIG. 6 is a diagram illustrating an example of traces of a PCB in accordance with various embodiments.

(7) FIG. 7 is a graph showing a comparison of thermal test data of the electric field shielding system in accordance with various embodiment herein and a conventional EMI shielding system.

(8) FIGS. 8A-8B are graphs showing a comparison of EMI results of a conventional EMI shielding system and the electric field shielding system in accordance with various embodiment herein.

DETAILED DESCRIPTION

(9) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the presented concepts. The presented concepts may be practiced without some or all of these specific details. In other instances, well known process operations have not been described in detail so as not to unnecessarily obscure the described concepts. While some concepts will be described in conjunction with the specific examples, it will be understood that these examples are not intended to be limiting.

(10) Embodiments of the disclosure provide an interface surface design for optimally addressing the EMI and thermal aspects of the system design. The embodiments described herein may use a split shield approach for the transmitter system interface surface design, which also serves the purpose of air channel on the interface surface. The embodiments described herein also provide an effective shield to electric field to reduce the EMI while providing effective air flow from the cooling fan to both the transmitter and receiver coils. The embodiments described herein can be used for different arrangements of the coils, such as horizontal placement and vertical placement. The embodiments described herein further address both the design aspects for receiver system design also by using a thermally conductive electric field shield. The embodiments described herein provide an innovative method that reliably detects the FO by using an eddy current induced in the electric field shield patterns. The embodiments described herein provide an ease of manufacturing by providing a unified interface surface with integration of an electric field shield

with thermal design, thereby eliminating the need of additional vents, befall for thermal aspects. Embodiments of the split shield described herein can be realized using a combination of any insulating and/or conductive materials, for example, plastic with conductive ink, flexible printed circuit board (PCB), etc. The embodiments described herein can be scaled and used for higher power (e.g., 15 W to <100 W) where EMI and thermal becomes even more important. The embodiments described herein can also be used for automotive applications where coils are encapsulated, and be used for different coil positions, such as horizontal and vertical/standing positions.

(11) In one aspect, a wireless power transmitter is provided. The transmitter may include a transmitter coil and a transmitter electric field shield disposed over the transmitter coil. The transmitter electric field shield may include a first printed circuit board (PCB) and a second PCB. The first PCB may be disposed over the second PCB. Each of the first and second PCBs may include a number of apertures. In an embodiment, the apertures of the first PCB do not overlap with the apertures of the second PCB.

(12) In another aspect, a wireless power receiver is provided. The receiver may include a receiver coil and a receiver electric field shield disposed over or underneath the receiver coil. The receiver electric field shield may include a PCB having a first layer and a second layer connected to one another. The first layer of the PCB may mate with the receiver coil and the second layer of the PCB may mate with an interface surface of the wireless power receiver.

(13) In yet another aspect, a wireless power transfer system is provided. The system may include a wireless power transmitter including a transmitter coil and a transmitter electric field shield disposed over the transmitter coil. The transmitter electric field shield may include a first PCB and a second PCB. The first PCB may be disposed over the second PCB. Each of the first and second PCBs may include a number of apertures. The apertures of the first PCB do not overlap with the apertures of the second PCB. The system may further include a wireless power receiver including a receiver coil and a receiver electric field shield disposed over or underneath the receiver coil. The receiver electric field shield may include a PCB having a first layer and a second layer connected to one another. The first layer of the PCB may mate with the receiver coil and the second layer of the PCB may mate with an interface surface of the wireless power receiver.

(14) FIG. 1 illustrates a diagram of an example of a system for wireless charging, configured in accordance with various embodiments. More specifically, a system, such as wireless charging system **100**, may be configured to support different operational modes for charging operations. As will be discussed in greater detail below, various components of wireless charging system **100** may include control logic that may be configured to determine transmission parameters used for wireless charging, and to determine which operational mode should be used at what time. In some embodiments, such management of different operational modes is performed using a single pulse width modulator, thus allowing implementation of such operational mode control without additional synchronization logic.

(15) In various embodiments, wireless charging system **100** includes Universal Serial Bus (USB) power adaptor **102**, wireless charging station **104**, and USB-Power Delivery (PD) integrated circuit (IC) controller **106** coupled to both USB power adaptor **102** and wireless charging station **104**. USB power adaptor **102** interfaces with a power source such as AC mains and outputs a voltage 'VIN' based on the power source. In some embodiments, USB power adaptor **102** plugs into wall outlet **108**. However, other AC or DC power source configurations are possible, such as for example a DC power source provided from a car battery. USB power adaptor **102** may be compliant with the USB-PD specification, USB-C specification, PPS (Programmable Power Supply) specification, etc. In general, the voltage VIN output by USB power adaptor **102** may have relatively small output voltage steps, e.g., every 10 mV, 40 mV, 100 mV, etc., or larger steps, e.g., 5V, 12V and 15V.

(16) Wireless charging station **104** wirelessly charges wireless charging device **110** such as a

cellular phone, smartphone, PDA (personal digital assistant), PDA phone, etc. in charging proximity of wireless charging station **104**. Wireless charging station **104** may be integrated in charging pad **112** and may include Wireless Power Inverter (WPI) **114** for wirelessly transferring power via magnetic induction to charge a battery included in wireless charging device **110** placed on charging pad **112**. WPI **114** may be a full-bridge or half-bridge inverter having voltage 'VBRG' as a DC input voltage, for example.

(17) Wireless charging station **104** includes an induction coil L_p placed in a series resonant circuit with a capacitor C_p to yield a resonant circuit with a natural resonance when coupled to the corresponding coil (not shown) included in wireless charging device **110**. When wireless charging device **110** is placed on charging pad **112**, the proximity of the coils allows an electromagnetic field to be created. This electromagnetic field allows power to pass from the coil L_p in charging pad **112** to the coil in wireless charging device **110**. The induction coil in wireless charging device **110** uses the transferred power to charge the device battery. More than one coil may be used on the transmit and receive sides.

(18) The same USB-PD IC controller **106** is used to control both wireless charging station **104** and USB power adaptor **102**. USB-PD IC controller **106** includes first USB port **116** for coupling USB-PD IC controller **106** to USB power adaptor **102** over USB cable **118**. USB-PD IC controller **106** may control USB power adaptor **102** via D+ and D- data pins on USB power adaptor **102**.

(19) USB-PD IC controller **106** also includes second port **120** for coupling the USBPD IC controller **106** to wireless charging station **104**. USB-PD IC controller **106** may control wireless charging station **104** via gate drive signal **122** provided to WPI **114** of wireless charging station **104**. For example, gate drive signal **122** may be a PWM (pulse width modulation) signal provided to a gate driver of WPI **114** for controlling the gates of power transistors that form full-bridge or half-bridge inverter of WPI **114**. USB-PD IC controller **106** may control wireless charging station **104** based on voltage and/or current information **124** received from wireless charging station **104**.

(20) USB-PD IC controller **106** also includes logic **126** for controlling the level of the voltage V_{IN} output by USB power adaptor **102** and the output power level of wireless charging station **104**. The input voltage 'VBRG' of wireless charging station **104** corresponds to the voltage V_{IN} output by USB power adaptor **102** or is derived from the voltage V_{IN} output by USB power adaptor **102**. As explained above, the USB power adaptor output voltage V_{IN} may have relatively small voltage steps, e.g., every 10 mV, 40 mV, 100 mV, etc.

(21) If the degree of voltage control available at USB power adaptor **102** is sufficient to implement the full output power range of wireless charging station **104**, the USB power adaptor output voltage V_{IN} may be input directly as the wireless charging station input voltage VBRG and USB-PD IC controller **106** may control the output power level of wireless charging station **104** by changing the level of V_{IN} and/or the operating frequency or duty cycle of wireless charging station **104**. If more granular voltage level control is needed to implement the full output power range of wireless charging station **104**, wireless charging system **100** may also include voltage regulator **128** such as a DC/DC switching regulator such as a buck regulator or other type of step-down converter for regulating the input voltage VBRG of wireless charging station **104** based on the voltage V_{IN} output by USB power adaptor **102**. In this case, USB-PD IC controller **106** also controls voltage regulator **128**, e.g., via gate drive signal **130** such as a PWM signal for controlling power transistors of voltage regulator **128**.

(22) FIG. 2 illustrates a diagram of another example of a system for wireless charging, configured in accordance with various embodiments. As similarly discussed above, a system, such as wireless charging system **200**, may be configured to support different operational modes for charging operations. Various components of wireless charging system **200** may include control logic that may be configured to determine transmission parameters used for wireless charging, and to determine which operational mode should be used at what time

(23) As similarly discussed above, system **200** may include a controller, such as controller **202**,

which is configured to control operation of various components within system **200**. More specifically, controller **202** may include components such as inverter controller **204** and pulse width modulator **208**. As similarly discussed above, inverter controller **204** may be configured to control operation of an inverter stage of a power inverter, such as inverter stage **212**. As will be discussed in greater detail below, inverter controller **204** may be configured to control input signals, such as input currents, provided to transistors included in inverter stage **212**. Accordingly, inverter controller **204** may be configured to control the operation of one or more transistors included within inverter stage **212**. In some embodiments, inverter stage **212** includes transistors configured as a half-bridge inverter. In various embodiments, inverter stage **212** includes transistors configured as a full-bridge inverter. As shown in FIG. 2, inverter stage **212** may be coupled to a transmission element, such as LC transmission element **214**, which is configured to transmit an output signal for wireless charging.

(24) System **200** further includes pulse width modulator **208** which is configured to generate a control signal used to drive power transfer operations. Accordingly, a control signal generated by pulse width modulator **208** may have a designated frequency, amplitude, and duty cycle. Thus, pulse width modulator **208** may be configured to generate a control signal that is ultimately used to drive other system components, such as inverter stage **212** and LC transmission element **214**. In various embodiments, one or more other components, such as logic device **210**, may generate additional signals based on an output of pulse width modulator **208**, and such additional signals may also be used for these purposes.

(25) System **200** additionally includes logic device **210** which is configured to receive a signal from pulse width modulator **208**, and generate one or more output signals based on the received signal. Logic device **210** is configured to generate multiple output signals based on a single signal received from a single pulse width modulator. Accordingly, multiple outputs may be generated by logic device **210**, and may be provided to components of controller **202**, such as inverter controller **204** to control components of inverter stage **212**, such as one or more transistors.

(26) System **200** may also include power adapter **206** which is configured to receive power from a power source. Accordingly, power adapter **206** may include a power sink and low-dropout (LDO) regulator configured to receive power via a cable, such as a USB cable, and provide power to other components of controller **202**.

(27) FIGS. 3A-3B are diagrams illustrating an example of a transmitter electric field shield in accordance with various embodiments. In some embodiments, the transmitter electric field shield may be integrated in wireless charging station **104** of FIG. 1 or wireless charging system **200** of FIG. 2. Referring to FIG. 3A, transmitter electric field shield **300** uses a split shield approach for an interface surface of a wireless power transmitter to shield electric field generated by the transmitter while allowing magnetic field to pass. As shown, shield **300** may be split between a first PCB **310** and a second PCB **311**. In addition to the shielding, each of the PCBs **310-311** also includes a number of apertures (or cut outs), for example apertures **320** and apertures **321** respectively, for cooling air to pass through. In this embodiment, the respective apertures **320-321** on both PCBs **310-311** are not overlapping. In an embodiment, PCB **310** and PCB **311** are disposed over one another, over a transmitter coil of the transmitter (not shown) to realize the interface surface of the wireless power transmitting system. In an embodiment, PCB **310** and PCB **311** may be in contact with one another, though in another embodiment, there may be a space in between them.

(28) With continued reference to FIG. 3A, PCB **311** may include traces **330** made of an electrically conductive material (e.g., copper traces). Traces **330** may be formed with specific patterns (e.g., open ended patterns) to produce an effective electric field shield. Referring now to FIG. 3B, the PCBs **310-311**, which may be disposed or formed at an interface surface of the transmitter, are assembled forming shield **300**. The interface surface formed with the assembled PCBs provides an uninterrupted electric field shielding for shield **300** as the apertures **320-321** on the assembled PCBs **310-311** are not overlapping. At the same time, the apertures **320-321** are configured such

that when both PCBs **310-311** are disposed over one another, they may provide an inlet path for air to flow to the interface surface, thereby creating a channel for air to flow over the interface surface and exit path. Accordingly, shield **300** addresses both cooling of the interface surface and reduction of EMI.

(29) FIG. **4** is a diagram illustrating an example of a receiver electric field shield in accordance with various embodiments. In some embodiments, the receiver electric field shield may be integrated or implemented in wireless charging device **110** of FIG. **1**, such as a cellular phone, smartphone, PDA (personal digital assistant), PDA phone, etc. Referring to FIG. **4**, receiver electric field shield **400** uses a split shield approach for an interface surface of a wireless power receiver to shield electric field generated by the receiver while allowing magnetic field to pass. In system **400**, shield **400** may be split between a first layer **411a** and a second layer **411b** of a same PCB **410**. In an embodiment, parts or an entirety of the PCB **410** in layer **411a** may be connected to parts or an entirety of layer **411b** through thermal vias **420** to ensure effective thermal conductivity between both layers/sides **411a-b** of PCB **410**. When the shielded PCB **410** is implemented with a receiver coil of the receiver (not shown), one of the PCB layer/side (e.g., layer **411a/411b**) may mate with the receiver coil and another PCB layer/side (e.g., layer **411a/411b**) may mate with the interface surface. The PCB **410** may be disposed over or underneath the receiver coil. In an embodiment, the PCB layer/side facing the interface surface may come in contact with the transmitter surface air channel. Since both PCB layers **411a-b** have good thermal conductivity, the cooling is effective for other side of the PCB layer/side which in turn cools the receiver coil. Also, the shield on both layers **411a-b** may block the electric field to reduce EMI. For example, one or both of the layers **411a-b** may include traces (e.g., traces **330** of FIG. **3** or other suitable traces) served to block the electric field produced by the receiver coil.

(30) FIG. **5** is a diagram illustrating an example of an electric field shielding system for interface surface cooling and shielding in accordance with various embodiments. Referring to FIG. **5**, electric field shielding system **500** includes, but not limited to, a wireless power receiver **501** and a wireless power transmitter **502**. As shown, wireless power receiver **501** includes, but not limited to, a receiver coil **510** and a receiver electric field shield **511**. The receiver electric field shield **511** may be disposed over or underneath the receiver coil **510** to shield or block the electric field produced by the receiver coil **510**, thereby effectively reducing EMI. As previously described, shield **511** may be formed with a PCB (e.g., PCB **410** of FIG. **4**) having a first layer/side and a second layer/side connected to one another through thermal vias for good thermal conductivity. In an embodiment, the first layer of the PCB may mate with (or be connected to) the receiver coil **510**. The second layer of the PCB may mate with (or be connected to) an interface surface **560** of the wireless power receiver **501**.

(31) With continued reference to FIG. **5**, wireless power transmitter **502** includes, but not limited to, a transmitter electric field shield **512** and a transmitter coil **520**. The transmitter electric field shield **512** may be disposed over the transmitter coil **520**. As shown, shield **512** may include a first PCB **513** and a second PCB **514**, with the PCB **513** being disposed over the PCB **514**. In an embodiment, PCBs **513-514** may be in contact with one another, though in another embodiment, there may be a space or gap in between them. In an embodiment, each of the PCBs **513-514** may include apertures (not shown) for cooling air (e.g., cooling air generated by fan **540**) to pass through. The apertures of PCBs **513-514** may be non-overlapping with one another when the PCBs **513-514** are assembled. Although not shown, PCB **514** may include traces made of an electrically conductive material, such as copper traces. As discussed in more detail herein below, the traces may be formed with specific patterns (e.g., open ended patterns) to produce an electric field shield. In an embodiment, shield **512** is placed over the transmitter coil **520** at the interface surface **561** of the wireless power transmitter **502** to shield the electric field produced by the transmitter coil **520** while allowing the magnetic field to pass. This therefore effectively reduces EMI, as shown in FIGS. **8A-8B**.

(32) Still referring to FIG. 5, system **500** may further include a support PCB **530** disposed underneath the transmitter coil **520**. Support PCB **530** may be seen as a “dummy” PCB that provides support for the transmitter coil **520**. In an embodiment, system **500** may also include a fan **540**. Fan **540** may generate cooling air through air channel **550** to cool the wireless power receiver **501** and the wireless power transmitter **502**. For example, air channel **550** may extend from fan **540**, through an inlet path of the transmitter coil **520** and the apertures of PCBs **513-514**, over the interface surface **561**, and to an exit path of the transmitter coil **520**. Furthermore, since the PCB layer/side of the receiver shield **511** that faces the interface surface **560** may come in contact with the air channel **550** at interface surface **561** and both PCB layers of receiver shield **511** have good thermal conductivity, the cooling enabled by the air channel **550** is effectively passed to the other PCB layer/side of the receiver shield **511**, which in turn cools the receiver coil **510**. Therefore, system **500** addresses both cooling of the receiver **501** and transmitter **502**, and reduction of EMI produced by coils **510** and **520**.

(33) FIG. 6 is a diagram illustrating an example of traces of a PCB in accordance with various embodiments. In some embodiments, traces **610** may be implemented on PCB **311** of FIG. 3A or PCB **514** of FIG. 5. Referring to FIG. 6, an interface surface (of a transmitter or receiver) may include traces **610** to form an electric field shield. The traces **610** may be formed using an electrically conductive material (e.g., copper traces). Traces **610** may have open-ended patterns which does not result in significant eddy currents. For example, when a foreign object **621** is placed on the interface surface, it may short a pair of adjacent traces and form an eddy current loop (e.g., eddy current loop **620**). In some embodiments, the eddy current loop **620** reduces the resistivity of the circuit and brings down the Q factor measured by the transmitter. Also, while the wireless power transfer system is in power delivery, the eddy current loop **620** may increase the system power loss.

(34) In a normal/conventional interface surface, presence of a foreign object may reduce the Q factor or increase the power loss of the system, but the change in those parameters may depend on the size/resistivity of the foreign object, and also the placement of the foreign object on the interface surface. In some operational contexts, there is a possibility that Q factor method or power loss method of FOD may not detect the foreign object, and thus, it can heat the system due to self-heating. However, in accordance with the techniques described herein, using the traces **610** as shown in FIG. 6, can help bring additional eddy current loss, thereby reducing the Q factor or increasing power loss significantly to make the foreign object detection reliable even when the foreign object is present at an undetected location, such as a corner of interface surface. It should be noted that the traces **610** illustrated in FIG. 6 are merely an example and other trace patterns may be utilized to achieve the same purpose.

(35) FIG. 7 is a graph showing a comparison of thermal test data of the electric field shielding system and a conventional EMI shielding system, in accordance with some embodiments herein. Referring to FIG. 7, in graph **700**, temperature **702** of a receiver coil assembled with the receiver electric field shield according to the embodiments herein, temperature **703** of a transmitter coil assembled with the transmitter electric field shield according to the embodiments herein, temperature **704** of a transmitter coil implemented with a conventional EMI shield, and temperature **705** of a receiver coil implemented with a conventional EMI shield are shown relative to ambient temperature **701**. As shown, the temperature **702** of the receiver coil assembled with the receiver electric field shield described herein is significantly lower than the temperature **705** of the receiver coil implemented with the conventional EMI shield. Correspondingly, the temperature **703** of the transmitter coil assembled with the transmitter electric field shield described herein is significantly lower than the temperature **704** of the transmitter coil implemented with the conventional EMI shield. Therefore, the receiver and transmitter electric field shields described herein are much more effective in cooling the receiver and transmitter coils than the conventional EMI shields.

(36) FIGS. 8A-8B are graphs showing a comparison of EMI results of a conventional EMI

shielding system and the electric field shielding system described herein. As illustrated in FIGS. 8A-8B, the EMI reduction of a wireless power transfer system having the receiver and transmitter electric field shields described herein (FIG. 8A) is significantly greater than the EMI reduction of a wireless power transfer system having conventional EMI shields (FIG. 8B). Therefore, the receiver and transmitter electric field shields described herein are also much more effective in reducing EMI as compared to the conventional EMI shields.

(37) In the foregoing specification, embodiments of the disclosure have been described with reference to specific example embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the invention as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A wireless power transfer system, comprising: a wireless power transmitter comprising a transmitter coil and a transmitter electric field shield disposed over the transmitter coil, the transmitter electric field shield comprising a first printed circuit board (PCB) and a second PCB, wherein the first PCB is disposed over the second PCB with a space therebetween, each of the first and second PCBs includes a plurality of apertures, and the apertures of the first PCB do not overlap with the apertures of the second PCB; and a wireless power receiver comprising a receiver coil and a receiver electric field shield disposed underneath the receiver coil, the receiver electric field shield comprising a PCB having a first layer and a second layer connected to one another, wherein the first layer of the PCB mates with the receiver coil and the second layer of the PCB mates with an interface surface of the wireless power receiver.
 2. The wireless power transfer system of claim 1, wherein the second PCB of the transmitter electric field shield further includes a plurality of traces to block electric field generated by the transmitter coil and allow magnetic field generated by the transmitter coil to pass.
 3. The wireless power transfer system of claim 2, wherein the traces have open-ended patterns.
 4. The wireless power transfer system of claim 2, wherein the traces are made of an electrically conductive material.
 5. The wireless power transfer system of claim 1, wherein the first layer of the PCB is connected to the second layer of the PCB through thermal vias.
 6. The wireless power transfer system of claim 2, further comprising a fan configured to generate cooling air that passes through an air channel, wherein the air channel extends through an inlet path of the transmitter coil, the apertures of the first PCB and the apertures of the second PCB, over an interface surface of the wireless power transmitter, and to an exit path of the transmitter coil.
 7. The wireless power transfer system of claim 6, wherein the interface surface of the wireless power receiver is in contact with the air channel to cool the receiver coil through thermal conductivity between the first layer of the PCB and the second layer of the PCB.
 8. The wireless power transfer system of claim 1, wherein the transmitter electric field shield is disposed at an interface surface of the wireless power transmitter.
 9. The wireless power transfer system of claim 1, wherein the receiver electric field shield is disposed at the interface surface of the wireless power receiver.
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